



AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Project Domain

[Excel / PowerBIwithCopilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



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Guide

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EMPLOYEE PAYROLL DATA CLEANING AND FINANCIAL ANALYSIS

PROBLEM STATEMENT

Organizations rely heavily on payroll systems to manage employee salaries, bonuses, and tax deductions. However, payroll data is often affected by inconsistencies such as missing values, duplicate records, incorrect data formats, and calculation errors. These issues can lead to inaccurate salary processing, financial mismanagement, and poor decision-making.

In many organizations, payroll datasets are large and complex, making manual verification difficult and time-consuming. Errors such as incorrect tax deductions or duplicate employee records may go unnoticed, leading to financial discrepancies.

Additionally, while Artificial Intelligence tools are becoming widely available, many users lack the skills to utilize them effectively for data analysis and reporting. This creates a gap between data availability and actionable insights.

This project aims to address these challenges by performing data cleaning and financial analysis on an employee payroll dataset. By applying Excel-based techniques and AI-assisted insights, the project demonstrates how structured data processing can improve accuracy, efficiency, and decision-making in payroll management.

OBJECTIVES

- To collect and explore a payroll dataset containing employee salary details.
- To identify and correct errors such as missing values, duplicates, and incorrect data formats.
- To perform financial analysis on employee salaries using Excel formulas and pivot tables.
- To create visual representations such as charts and dashboards for better understanding.
- To use an AI chatbot to generate insights and summaries from the dataset.
- To improve data accuracy and demonstrate efficient payroll data handling techniques.

METHODOLOGY / IMPLEMENTATION PLAN

#	Phase	Description
1	Data Collection & Exploration	Obtain the raw dataset (CSV / Excel) from Kaggle, Edunet, or any suitable source. Understand field names, data types, and row count. For AI prompt projects, define the use case and target audience instead.
2	Data Cleaning & Preprocessing	Remove duplicates, handle null or missing values, fix incorrect data types, and standardize data and text formats using Excel (Remove Duplicates, IF, IFERROR, Text-to-Columns) or Power Query. For prompt-based projects, draft and structure initial prompts using Role, Context, Task, Format, and Tone components.
3	Analysis & AI Integration	Apply pivot tables, Excel formulas (SUMIF, COUNTIF, VLOOKUP), or Power BI DAX measures to derive key insights. Where applicable, use Microsoft Copilot or an AI Chatbot to generate narratives, summaries, or content outputs from the data or prompts.
4	Visualization / Output	Build interactive dashboards with charts, KPI cards, and slicers in Power BI or Excel. For AI-assisted projects, compile generated content outputs such as reports, marketing copy, or summaries. Document all AI prompts used alongside their outputs.
5	Documentation & Submission	Write the project report, upload all files to a GitHub repository, and record a video demonstration. Submit the GitHub link, video link, and dashboard link (if applicable) for evaluation.

- This project follows a systematic approach to clean and analyze employee payroll data using Microsoft Excel. Initially, a raw dataset containing errors such as missing values, duplicate records, and incorrect data formats was examined. Data cleaning techniques were applied to identify and correct these issues, including handling missing values, removing duplicates, and standardizing data types.
- After cleaning, formulas were used to calculate salary components, and pivot tables were created for analysis. Visualizations such as charts were used to represent the data clearly. Each step of the process was supported with screenshots to demonstrate the transformation from raw data to a structured and accurate dataset.

Phase 1: Data Collection & Exploration

- A payroll dataset was created to simulate real-world employee salary records. The dataset includes fields such as Employee ID, Name, Department, Basic Salary, Bonus, Tax, Net Salary, and Payment Date.
- Initial exploration revealed several inconsistencies such as missing values, incorrect data entries, and duplicate records.

Employee ID	Employee Name	Gender	Age	Department	Designation	Joining Date	Basic Salary	Allowances	Bonus	Tax Deductible	Net Salary	Taxance	Performance	Payment Date
1	EMP001	Bryan Dav	Male	Ops	Operation	22-05-2022	€81395	10834	11533	11956.27	91805.73	85	5	06-05-2026
2	EMP002	Michael Fi	Male	29 sales	Sales	Mar 2019/01/3	€81017	11770	4228	13469.62	83545.38		2	04/11/2026
4	EMP003	Daniel Da	FEMALE	Finance	Accountar	11/20/2022	59,632	13905	3102	4064.77	72574.23	100		04/15/2026
5	EMP004	Glenn Alv	FEMALE	HR	HR Execu	04-03-2022	€46657	14838	4003	12119.73	71378.27		4	2026/05/03
6	EMP005	Shawn Sa	FEMALE	MKT	SEO Anal	13-11-2022	€46087	8134	6419	5458.15	55181.85		5	04/22/2026
7		Evan Daw	male	Ops	Operation	2021/11/1	75035	6209	8355	11837.25	77761.75		1	2026/04/06
8	EMP007	Jenna Da	Female	hr	HR Manag	2021/05/0	€72709	3725	5085	7230.48	74288.52	92	5	2026/05/07
9	EMP008	Briana Ha	Male	sales	Sales	Mar 09-03-2022	78018	15647	9903	14739.1	88828.9		3	2026/03/31
10	EMP009	Daniel Thi	Male	I.T	System Ar	02/23/2022	77529		12756	14760.75	91380.25	75		2026/04/15
11	EMP010	Brian Smi	male	24 Operation	Operation	28-12-2022	28,953	5651	1141	5369.16	30115.84		4	2026/03/26
12	EMP011	Mary Burk	male	24 it	System Ar	2019/03/1	63295	11730	9747	10067.55	74704.35		4	2026/03/20
13	EMP012	Gary Garc	FEMALE	MKT	SEO Anal	06/27/2022	€89317	11548	11378	12027.22	100215.8		5	2026/04/06
14	EMP013	Robert He	male	HR	HR Execu	10/30/2022	83,934	15782	9372	17638.18	91449.82	86	3	2026/05/01
15	EMP014		FEMALE	MKT	Marketing	2017/04/1	€89276	8525	14455	15941.18	96314.82		5	2026/03/22
16	EMP015	Linda Gra	Male	Operation	Operation	04-05-2022	€57423	11560	4074	8022.36	75034.64		3	27-04-2026
17	EMP016	Ian Robbr	FEMALE	FINANCE	Accountar	10/31/2022	82,597	5830	1448	11082.28	78792.72	74		04/19/2026
18	EMP017	Danielle D	Female	26 SALES	Sales	Mar 06/08/2022	69,289	12836	1904	12218.02	71810.98	74	4	2026/04/12
19	EMP018	Joshua C	FEMALE	I.T	Software E	11/15/2022	42767	11681	8620	10328.28	52739.72		3	04/14/2026
20	EMP019	Tara Price	male	54 hr	HR Execu	08/25/2022	84,969	8585	12711	12772.02	93512.98	98	5	30-03-2026
21	EMP020	Austin Rey	Male	FINANCE	Accountar	05/18/2022	51611	16187	14796	6910.58	75683.42		2	19-04-2026

Figure 1: Data Collection (Uncleaned)

Phase 2: Data Cleaning & Preprocessing

The dataset was cleaned using various Excel tools and techniques to ensure accuracy and consistency.

1. Identifying Errors

Errors such as missing salary values, duplicate records, invalid text entries, and incorrect date formats were identified.

2. Removing Duplicates

Duplicate employee records were removed using Excel's "Remove Duplicates" feature.

3. Handling Missing Values

Missing salary values were filled using appropriate estimation methods.

4. Correcting Data Types

Text values in numeric columns were converted into a proper numeric format.

5. Fixing Invalid Values

Negative and unrealistic values such as incorrect tax entries were corrected.

6. Fixing Date Format

Incorrect and inconsistent dates were corrected.

7. Final Clean Dataset

After all corrections, the dataset became structured and ready for analysis.

Employee_Payroll_Cleaned_Dataset

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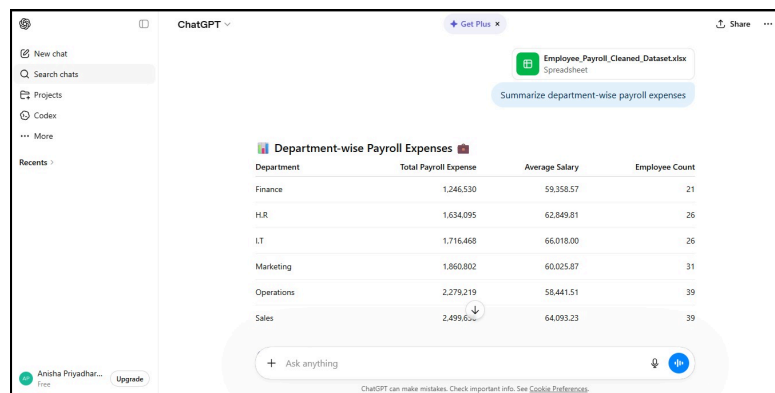
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Figure 2: Cleaned Dataset Version

Phase 3: Analysis & Ai Integration

After cleaning, financial analysis was performed to extract meaningful insights. Excel functions such as SUM, AVERAGE, and IF were used to calculate salary-related metrics. Pivot tables were created to analyze department-wise salary distribution. for assisting with these tasks Ai integration helps. Example ; ChatGPT.



The screenshot shows the ChatGPT interface with a table titled "Department-wise Payroll Expenses". The table has four columns: Department, Total Payroll Expense, Average Salary, and Employee Count. The data is as follows:

Department	Total Payroll Expense	Average Salary	Employee Count
Finance	1,246,530	59,358.57	21
H.R	1,634,095	62,849.81	26
IT	1,716,468	66,018.00	26
Marketing	1,860,802	60,025.87	31
Operations	2,279,219	58,441.51	39
Sales	2,499,600	64,093.23	39

Figure 3: Ai Integration (ChatGPT)

Phase 4: Visualization / Output

Charts and graphs were created to visually represent payroll insights in the Dashboard.

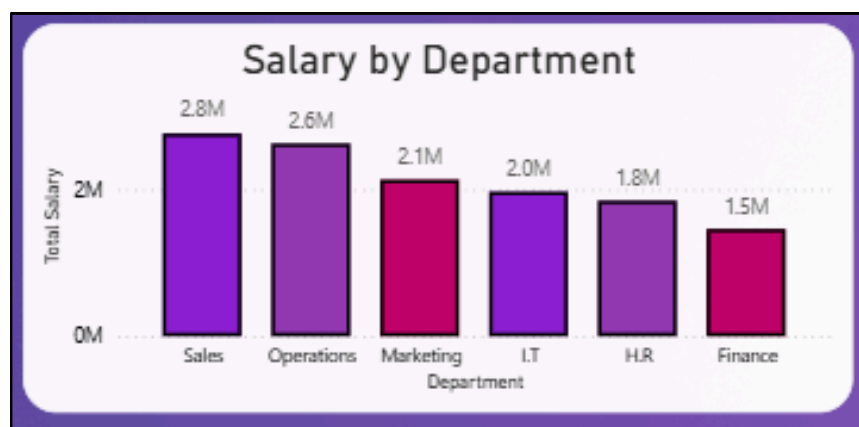


Figure 4: Bar Chart

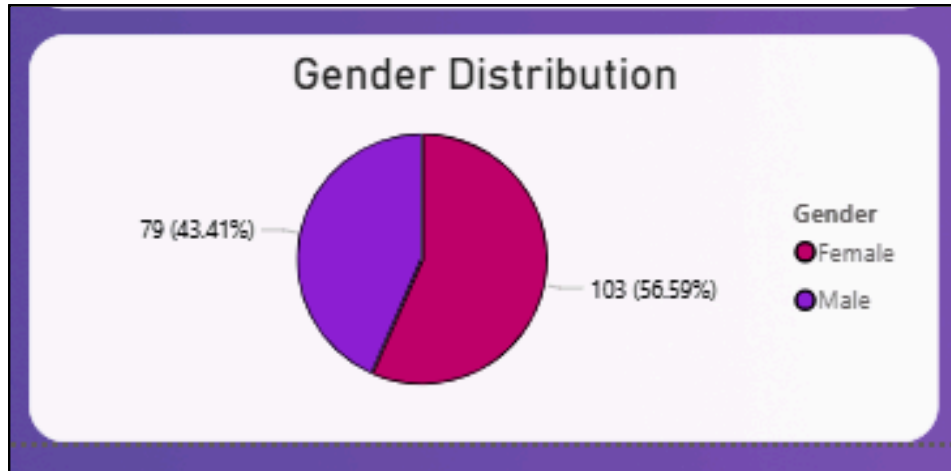


Figure 5: Pie Chart

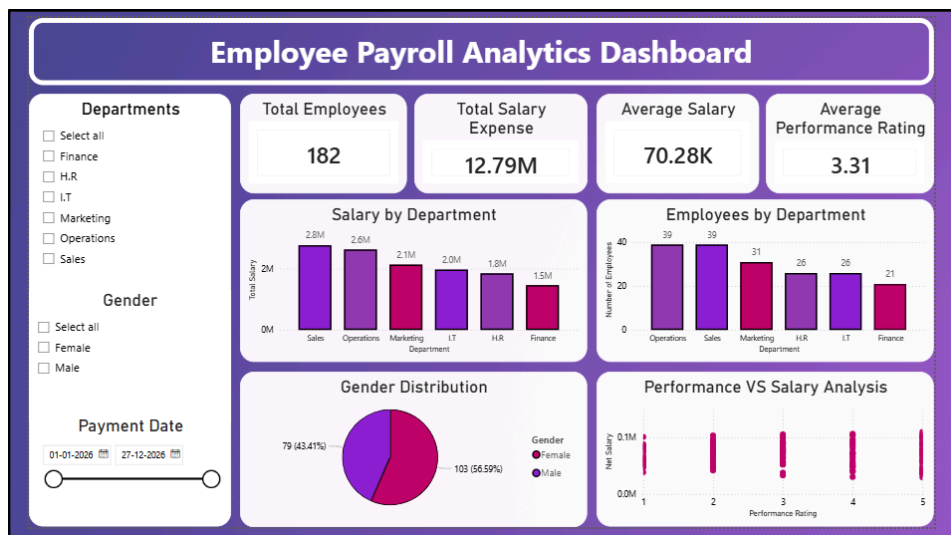


Figure 6: Employee Payroll Dashboard

Phase 5: Documentation & Submission

- All steps of the project were documented, and outputs were prepared for submission including dataset, analysis, and report.
- The cleaned dataset and analysis results were verified to ensure accuracy and consistency before submission.

RESOURCES AND TIMELINE

A. Hardware & Software Requirements

- Laptop with minimum 4 GB RAM
- Microsoft Excel (2019 or later)
- AI Chatbot (ChatGPT)
- Internet connection
- Web browser

B. Dataset

- The dataset used in this project is a simulated payroll dataset created to represent real-world employee salary records. It includes details such as employee information, salary components, tax deductions, and payment dates.
- The dataset was intentionally designed with errors such as missing values, duplicates, and incorrect formats to demonstrate the data cleaning process.

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	<i>Week 1</i>
2	Data cleaning, preprocessing, and validation / Prompt design	<i>Week 2</i>
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	<i>Week 2 – 3</i>
4	Dashboard / output build, testing, and review	<i>Week 3</i>
5	Report writing, GitHub upload, and video recording and submission	<i>Week 4</i>

EXPECTED OUTCOMES / CONCLUSION

The project is expected to result in a clean and structured payroll dataset with all inconsistencies removed. Financial analysis will provide insights into salary distribution, department-wise expenses, and employee compensation patterns.

The use of AI tools will enhance the ability to generate insights and summaries efficiently. The final output will include a well-documented report, visual charts, and a clear demonstration of data cleaning and analysis techniques.

This project highlights the importance of data accuracy and the role of data analytics in improving organizational decision-making.

REFERENCES

- Microsoft Excel Documentation
- Kaggle Dataset Resources
- Microsoft Copilot & ChatGPT (AI Chatbot)
- Online tutorials and learning materials

LINKS

- Microsoft Corporation. Microsoft Copilot. Available at: <https://copilot.microsoft.com>
- Kaggle Inc. Open Datasets. Available at: <https://www.kaggle.com/datasets>
- Open AI, ChatGPT. Available at: <https://chatgpt.com/>
- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: <https://www.edunetfoundation.org>
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: <https://www.linkedin.com/learning>
- Microsoft Corporation. Microsoft Excel Documentation. Available at: <https://support.microsoft.com/en-us/excel>
- Microsoft Corporation. Power BI Documentation. Available at: <https://learn.microsoft.com/en-us/power-bi/>
- Github, Project Available at : <https://github.com/anishapriyadharshini/Employee-Payroll-Data-Cleaning-And-Financial-Analysis-.git>

Thank You